

C784

Applied Healthcare Statistics

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Lengths of stay on a cardiac unit, which was renovated last year, for seven recent patients were 2, 3, 3, 4, 4, 5 and 30 days. Which measure best describes a typical stay?

- A. The range, 28 days
- B. The median, 4 days
- C. The mean, about 7.3 days
- D. The mean, since it is always the best measure

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

In a large screening program, total cholesterol has a mean of 200 mg/dL and a standard deviation of 25. A 54-year-old patient who exercises three times a week has a reading of 250. What is the patient's z-score?

- A. 50
- B. 10.0
- C. 0.5
- D. 2.0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Resting heart rates have a mean of 120 (in a pediatric group) and a standard deviation of 10. A child's z-score is -1.5 . What is the child's heart rate?

- A. 15
- B. 135
- C. 105
- D. 118.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Using data from a state health department website updated monthly, an analyst finds $r = 0.82$ between the number of hospitals in a county and the number of deaths in that county. What is the best conclusion?

- A. Hospitals cause deaths, so closing some would save lives.
- B. There is a strong positive association, likely driven by a lurking variable such as population.
- C. 82% of deaths are explained by hospital count.
- D. There is a strong negative association.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A model built from 2,000 patient records predicts systolic blood pressure as $\hat{y} = 90 + 0.6x$, where x is age in years. A 50-year-old patient weighs 180 pounds. What systolic pressure does the model predict?

- A. 4,530 mmHg
- B. 120 mmHg
- C. 30 mmHg
- D. 90 mmHg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Using $\hat{y} = 90 + 0.6x$ (x = age), the model predicts a systolic pressure of 132. What age does that correspond to?

- A. 42 years
- B. 220 years
- C. 370 years
- D. 70 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A clinic that is open six days a week has 200 patients with an average age of 61. Sixty have diabetes, 80 have hypertension, and 30 have both. What is the probability a randomly chosen patient has diabetes or hypertension?

- A. 0.55
- B. 0.15
- C. 0.45
- D. 0.70

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

$P(A) = 0.5$, $P(B) = 0.4$ and $P(A \text{ or } B) = 0.7$. What is $P(A \text{ and } B)$?

- A. 0.30
- B. 0.70
- C. 0.20
- D. 0.90

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A screening test, which costs \$12 per kit, was given to 1,000 patients. Of 100 with the disease, 90 tested positive. Of 900 without it, 45 tested positive. If a patient tests positive, what is the probability the patient has the disease?

- A. 0.09
- B. 0.90
- C. About 0.67
- D. 0.135

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Of 900 patients without a disease, 855 tested negative; of 100 with it, 10 tested negative. What is the probability of a negative test given that a patient does not have the disease?

- A. 0.05
- B. 0.855
- C. About 0.99
- D. 0.95

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a study of 5,000 adults, those who chose to take vitamin D supplements had 20% fewer respiratory infections than those who did not. The difference was statistically significant. Can researchers conclude vitamin D prevents infections?

- A.** No; this is an observational study, and confounding (for example, healthier habits) could explain it.
- B.** Yes, because the result is statistically significant.
- C.** No, because no study can ever show causation.
- D.** Yes, because 5,000 people is a large sample.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A hospital wants to measure satisfaction among all patients discharged last month. The survey was designed by a consultant with 15 years of experience. Which sampling plan gives the most representative results?

- A.** Randomly select 300 patients from the full discharge list.
- B.** Survey every patient discharged from the orthopedic unit.
- C.** Post a link on the website for anyone who wants to respond.
- D.** Survey the first 300 people in the cafeteria at lunch.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

Lengths of stay on a cardiac unit, ~~which was renovated last year,~~ for seven recent patients were 2, 3, 3, 4, 4, 5 and 30 days. Which measure best describes a typical stay?

Target word: best measure of center **Decides:** The 30-day stay is an outlier.

A	Confusion set	Range measures spread, not center.
B	Correct	The middle of the ordered values is 4 and is not pulled by the 30.
C	Adjacent category	$51 / 7 \approx 7.3$ is pulled up by the outlier.
D	Absolute language	The mean is poor when data are skewed.

2. Answer D Concept 2

In a large screening program, total cholesterol has a mean of 200 mg/dL and a standard deviation of 25. A ~~54-year-old patient who exercises three times a week~~ has a reading of 250. What is the patient's z-score?

Target word: z-score **Decides:** $(250 - 200) / 25$.

A	Adjacent category	The raw difference, not standardized.
B	Dropped element	$250 / 25$ drops subtracting the mean.
C	Inverted operation	$25 / 50$ flips the ratio.
D	Correct	$50 / 25 = 2.0$.

3. Answer C Concept 2 · transfer

Resting heart rates have a mean of 120 (in a pediatric group) and a standard deviation of 10. A child's z-score is -1.5 . What is the child's heart rate?

Target word: solve for the value **Decides:** $x = \text{mean} + z \times \text{SD}$.

A	Adjacent category	The distance from the mean, not the heart rate.
B	Inverted operation	Ignores the negative sign and goes above the mean.
C	Correct	$120 + (-1.5 \times 10) = 105$.
D	Dropped element	Subtracts 1.5 without multiplying by the SD.

4. Answer B Concept 3

Using data ~~from a state health department website updated monthly,~~ an analyst finds $r = 0.82$ between the number of hospitals in a county and the number of deaths in that county. What is the best conclusion?

Target word: what r supports **Decides:** Correlation only; population size likely drives both.

A	Confusion set	Confuses association with causation.
B	Correct	Larger counties have more hospitals and more deaths; r does not show cause.
C	Wrong stem number	Misreads r as a percent; even r^2 (0.67) is not a causal share.
D	Inverted operation	A positive r means both rise together.

5. Answer B Concept 4

A model built from 2,000 patient records predicts systolic blood pressure as $\hat{y} = 90 + 0.6x$, where x is age in years. A 50-year-old patient weighs 180 pounds. What systolic pressure does the model predict?

Target word: predicted y **Decides:** $90 + 0.6 \times 50$.

A	Inverted operation	$(90 + 0.6) \times 50$ adds before multiplying.
B	Correct	$0.6 \times 50 = 30$; $90 + 30 = 120$.
C	Dropped element	Leaves out the intercept.
D	Wrong stem number	The intercept alone.

6. Answer D Concept 4 • transfer

Using $\hat{y} = 90 + 0.6x$ (x = age), the model predicts a systolic pressure of 132. What age does that correspond to?

Target word: solve for x **Decides:** $(132 - 90) / 0.6$.

A	Dropped element	Subtracts the intercept but forgets to divide by the slope.
B	Dropped element	$132 / 0.6$ ignores the intercept.
C	Inverted operation	$(132 + 90) / 0.6$ adds the intercept instead of subtracting.
D	Correct	$42 / 0.6 = 70$.

7. Answer A Concept 5

A clinic that is open six days a week has 200 patients with an average age of 61. Sixty have diabetes, 80 have hypertension, and 30 have both. What is the probability a randomly chosen patient has diabetes or hypertension?

Target word: P(A or B) **Decides:** Subtract the 30 counted twice.

A	Correct	$(60 + 80 - 30) / 200 = 110 / 200$.
B	Adjacent category	$30 / 200$ is P(both), not P(either).
C	Inverted operation	$90 / 200$ is P(neither), the complement.
D	Dropped element	$140 / 200$ counts the overlap twice.

8. Answer C Concept 5 • transfer

$P(A) = 0.5$, $P(B) = 0.4$ and $P(A \text{ or } B) = 0.7$. What is $P(A \text{ and } B)$?

Target word: solve for the overlap **Decides:** $0.5 + 0.4 - P(A \text{ and } B) = 0.7$.

A	Adjacent category	$1 - 0.7$ is P(neither).
B	Wrong stem number	P(A or B) from the stem.
C	Correct	$0.9 - 0.7 = 0.2$.
D	Inverted operation	Adds the probabilities and skips the subtraction.

9. Answer C Concept 6

A screening test, ~~which costs \$12 per kit~~, was given to 1,000 patients. Of 100 with the disease, 90 tested positive. Of 900 without it, 45 tested positive. If a patient tests positive, what is the probability the patient has the disease?

Target word: P(disease | positive) **Decides:** Denominator = all positives = 90 + 45.

A	Dropped element	90 / 1,000 drops the condition and uses everyone.
B	Inverted operation	90 / 100 is P(positive disease), the reverse condition.
C	Correct	90 / 135 $\approx$ 0.67.
D	Adjacent category	135 / 1,000 is P(positive).

10. Answer D Concept 6 · transfer

Of 900 patients without a disease, 855 tested negative; of 100 with it, 10 tested negative. What is the probability of a negative test given that a patient does not have the disease?

Target word: P(negative | no disease), a different conditional **Decides:** Denominator = 900 without the disease.

A	Adjacent category	45 / 900 is the false-positive rate.
B	Dropped element	855 / 1,000 uses everyone instead of the condition.
C	Inverted operation	855 / 865 is P(no disease negative), the reverse.
D	Correct	855 / 900 = 0.95.

11. Answer A Concept 7

In a study ~~of 5,000 adults~~, those who chose to take vitamin D supplements had 20% fewer respiratory infections than those who did not. The difference was statistically significant. Can researchers conclude vitamin D prevents infections?

Target word: causal claim allowed? **Decides:** Participants chose their own treatment; no random assignment.

A	Correct	Without random assignment, only association is shown.
B	Adjacent category	Significance shows the difference is unlikely by chance, not what caused it.
C	Absolute language	Randomized experiments can support causal claims.
D	Camouflage	Size does not remove confounding.

12. Answer A Concept 8

A hospital wants to measure satisfaction among all patients discharged last month. ~~The survey was designed by a consultant with 15 years of experience.~~ Which sampling plan gives the most representative results?

Target word: representative sampling method **Decides:** Every discharged patient needs an equal chance of selection.

A	Correct	A simple random sample from the whole population avoids selection bias.
B	Dropped element	Drops every other unit from the population.
C	Confusion set	Voluntary response draws people with strong opinions.
D	Adjacent category	A convenience sample; visitors and staff are not the population.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	9	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	3	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Mean vs. median with outliers	The median resists outliers; with skewed data it is the better measure of center.	Always reporting the mean because it uses every value.	One value far from the rest, such as a very long hospital stay.
2	z-scores	$z = (\text{value} - \text{mean}) / \text{standard deviation}$; it counts standard deviations from the mean.	Dividing the raw value by the SD, or flipping the sign.	A mean, a standard deviation, and one patient's value.
3	Correlation is not causation	r measures the strength and direction of a linear association; it cannot show cause, and lurking variables are common.	Reading a strong r as proof that one variable causes the other.	Two variables that both grow with population or size.
4	Using a regression line	$\hat{y} = \text{intercept} + \text{slope} \times x$; the slope is the predicted change in y for each one-unit change in x.	Dropping the intercept or doing operations out of order.	An equation of the form $\hat{y} = a + bx$ and an x value.
5	Addition rule	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.	Adding without subtracting the overlap.	"Or" with groups that can overlap.
6	Conditional probability	$P(A B) = \text{count of A and B} / \text{count of B}$; the given condition sets the denominator.	Using the whole sample as the denominator, or reversing the condition.	"Given that", "among those who", or "if a patient tests positive".
7	Experiments vs. observational studies	Only random assignment to treatment supports cause-and-effect claims; observational studies show association.	Accepting causation because the sample is large or the result significant.	Participants chose their own treatment.
8	Sampling methods and bias	A simple random sample from the full population is representative; convenience and voluntary samples are biased.	Thinking a large convenience sample fixes bias.	How participants were picked: whoever was handy, whoever volunteered, or at random.

Course scope (public description): Applied Healthcare Probability and Statistics is designed to help develop competence in the fundamental concepts of basic mathematics, introductory algebra, and statistics and probability. These concepts include basic arithmetic with fractions and signed numbers; introductory algebra and graphing; descriptive statistics; regression and correlation; and probability. Statistical data and probability are now commonplace in the healthcare field. This course will help candidates make informed decisions about which studies and results are valid, which are not, and how those results affect your decisions. This course will give candidates background in what constitutes sound research design and how to appropriately model phenomena using statistical data. Additionally, this course guides candidates in calculating simple probabilities based on events which occur in the healthcare profession. This course will prepare candidates for studies at WGU, as well as in the healthcare profession.